

12 Ans: B

$$\frac{GM_E/r_E^2}{GM_M/r_M^2} = 6 \dots\dots\dots(1) \qquad \frac{\rho_E}{\rho_M} = \frac{M_E/(4/3)\pi r_E^3}{M_M/(4/3)\pi r_M^3} = \frac{5}{3} \dots\dots\dots(2)$$

From (1), $\frac{M_E}{M_M} = \frac{6r_E^2}{r_M^2} \dots\dots\dots(3)$

Substitute (3) into (2), $\frac{6r_E^2}{r_M^2} \times \frac{r_M^3}{r_E^3} = \frac{5}{3} \Rightarrow \frac{r_E}{r_M} = 3.6$

13 Ans: B